

How neural meets symbolic

A simple guide to this lab (and the online demo)

Short companion to the full progress notes — for readers new to neuro-symbolic AI.

Demo: <https://udonsikalu.github.io/cxr-evidence-grounding-lab/>

Full record: [notes/progress-notes.pdf](#)

Read this if you want the idea in plain language, then open the demo.
Read the full notes if you want every protocol number, held-out table, and freeze date.

1 The idea in one picture

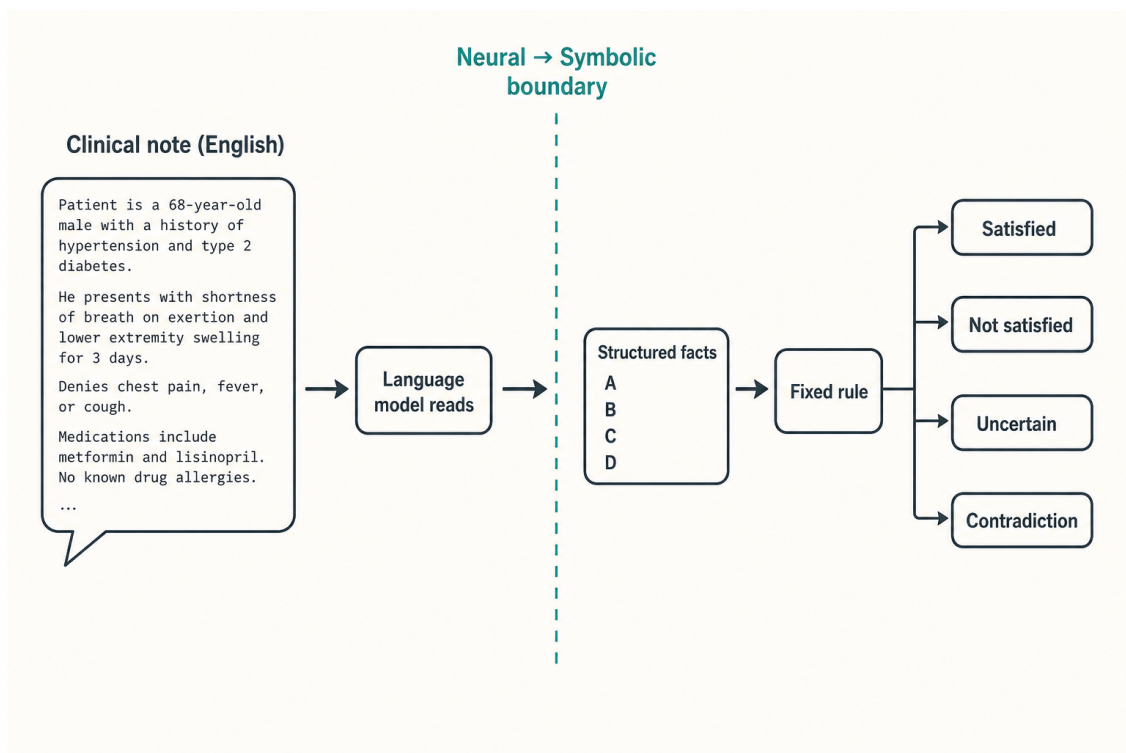


Figure 1: A language model reads the note. Structured facts cross a boundary into a fixed rule. The rule — not the model — issues the final label.

Neuro-symbolic means: neural (the model) + symbolic (clear rules). The interesting failure is not “the model is dumb.” It is: *the model seemed to understand, but the formal structure dropped the point.*

2 Our toy question

One made-up clinical question (synthetic notes, not real patients):

Did **first-line therapy** fail?

A fixed rule needs four facts (and a contradiction flag):

Atom	Plain meaning
A	A first-line therapy is named
B	It was actually given / tried
C	Something counts as failure
D	That failure is of <i>that</i> first-line regimen
X	The note contradicts itself about the above

Same atoms in $\rightarrow$ same verdict out: Satisfied, Not satisfied, Uncertain, or Contradiction.

3 Two ways to answer the same note

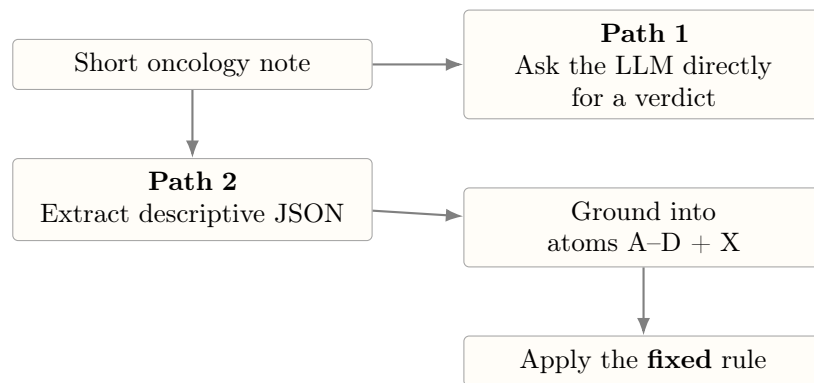


Figure 2: We always compare both paths on the same note. Gold labels are never shown to the model.

If Path 1 and Path 2 disagree, something was lost at the boundary.

4 What the online demo shows

The public page is **replay only**: it loads frozen experiment files. It does **not** run a model on anyone's computer.

N2S evidence grounding — findings demo
Artifact replay only — frozen Phase 1–4 results. No live models, no calls to anyone's GPU. Clone the repo to run Ollama locally on :8253 / :8254 .

ARTIFACT: Phase-4 held-out (C9-U12) MODEL: llama3.8b-instruct-q4_0 **Load** loaded phase4-heldout-panel.json · replay only (no inference)

Model: llama3.8b-instruct-q4_0 A 2/7 B 4/7 C 6/7 D 5/7 D rep-loss misses: 1

llama3.8b-instruct-q4_0 A2/7 mistral:instruct A7/7 qwen2.5-coder:32b A2/7

Case	Verdict	Match	Loss stage
A	NOT SATISFIED	no	direct_verdict_miss
B	UNCERTAIN	no	analysis_output_misalignment
C	CONTRADICTION	yes	none
D	CONTRADICTION	yes	none

C9 — GOLD CONTRADICTION - CONTRADICTION
 07:40 tumor-board list: first-line ipilimumab/nivolumab has failed; schedule second-line BRAF-directed therapy. 19:10 same-day clinic addendum: restaging today is stable disease on first-line ipilimumab/nivolumab; keep current immunotherapy, do not switch.
 Same-day reverse: morning note says first-line failed and switch; evening note says continue first-line with stability.

A — DIRECT VERDICT
 verdict: NOT SATISFIED ✗ gold
 loss: direct_verdict_miss
 The note states that the first-line therapy has failed and a second-line treatment is scheduled, but then it contradicts this by saying to continue the current immunotherapy without switching.

B — ANALYSIS — VERDICT
 verdict: UNCERTAIN ✗ gold
 loss: analysis_output_misalignment
 The note suggests that a first-line therapy is identified (ipilimumab/nivolumab), as explicitly stated in the tumor-board list at 07:40. This meets condition A.
 Regarding condition B, it appears that the first-line therapy was given or attempted, as the note mentions "first-line ipilimumab/nivolumab has failed", implying that the treatment was administered before its failure.
 The note also indicates a failure event (condition C) with the statement "first-line ipilimumab/nivolumab has failed" in the tumor-board list. This suggests that the patient's disease progressed or did not respond to the first-line therapy, meeting condition C.
 However, the addendum at 19:10 contradicts this notion, stating that restaging shows stable disease on the same first-line ipilimumab/nivolumab regimen. This implies that the failure event may not be a true progression or non-response, but rather a

Figure 3: GitHub Pages demo — pick an artifact and a model, then click cases in the left list. Live URL: <https://udonsikalu.github.io/cxr-evidence-grounding-lab/>

4.1 H

ow to walk the demo (2 minutes)

1. Open the link above (or open docs/ locally).
2. Keep the default **Phase-4** artifact, or try Phase-2 / Phase-3.
3. Click the model buttons (Llama / Mistral / Qwen) — same notes, different models.
4. Click a case on the left. Read the note, then the A–D table.

4.2 W

hat A–D mean in the demo

These are **not** four medical rules. They are four places meaning can disappear:

Col.	Plain meaning
A	Direct LLM verdict (no structure).
B	Model writes an analysis, then a final answer — does the answer keep what the analysis said?
C	Full pipeline: extract → atoms → fixed rule.
D	Analysis had the distinction, but the structured form dropped it (“representation loss”).

Loss stage under each column tells you *where* it broke, not just that it missed gold.

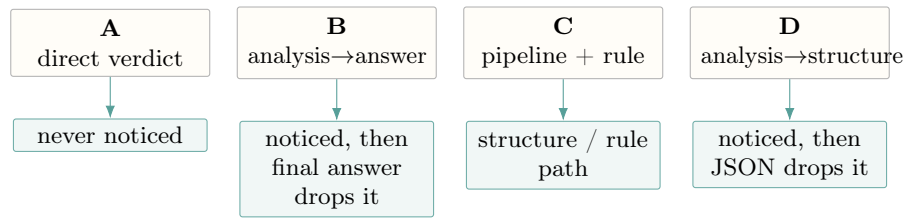


Figure 4: Conditions A–D map to different failure stories. That is why the demo shows all four.

5 The story in three beats

1. **Make the boundary visible.** Same notes, direct LLM vs structured pipeline (Milestone 1 UI on port 8253 if you run locally).
2. **Keep hard states on purpose.** Mark contradiction and uncertainty in the structured form so the rule can fire — states the raw LLM often narrates and then throws away.
3. **Ask where meaning died.** Conditions A–D on discovery notes, then held-out notes, across three local models. Representation loss (D) still appeared on unseen wording.

That is the arc. The long PDF lists every count and protocol freeze.

6 What we are (and are not) claiming

- **Are:** a small synthetic lab that makes the neural→symbolic handoff *inspectable*.
- **Are not:** a clinical product, a medical device, or the full CXR stack (no Qdrant / Archetypes here).
- **Public demo:** frozen files only — safe to share; no inference on the author’s servers.

7 Where to go next

Want...	Open...
Click around	https://udonsikalu.github.io/cxr-evidence-grounding-lab/notes/progress-notes.pdf
Every number & freeze	
Written journey	EVALUATION-JOURNEY.md
Run the local UI	python3 server.py → http://127.0.0.1:8253/
Repo README (NeSy intro)	GitHub project homepage

Rebuild: `pdflatex progress-notes-simple.tex` from the `notes/` directory.